SMe Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y} = (Y_1, ..., Y_n)$ Sample Data: $\mathbf{y} = (y_1, ..., y_n)$ $f_{\mathbf{Y}}(\mathbf{y};\theta) = \prod_i f_{\mathbf{Y}}(y_i;\theta)$,if y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \le y_1, ..., Y_n \le y_n)$
Parameter Space: Θ CDF: $F_Y(y)$ PDF/PMF: $f_Y(y)$ or $p_Y(y)$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i)\mathbb{P}(Y = y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y)f_Y(y)dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \subseteq \Theta\}$ is model.

Likelihood Function: $L(\theta;\mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y};\theta)$ ·**MLE**: $\hat{\theta}(\mathbf{y}) = ArgMax_{\theta\in\Theta}[L(\theta;\mathbf{y})] = [U(\hat{\theta}) = 0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta;x) \propto L(\theta;y)$ [Strong: $L_f(\theta;x) \propto L_g(\theta;z)$; f, g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y;\theta_1) = f(y;\theta_2), \forall y$ then $\theta_1 = \theta_2$ (i.e.$map: \theta \to f(y;\theta)$ is 1-1)

· Method: **1.**By def $\Rightarrow$ ratio not depend on $y \Rightarrow \theta_1, \theta_2$ relationship. (with any additional constraint) **2.**MLE not unique $\Rightarrow$ not identifiable **3.**Give an example θ **4.**Fisher information

Linear Algebra Knowledge: If A is symmetric matrix, with eigenvalues λ_n then $A + aI$ have eigenvalues $\lambda_n + a$. || $det(M) = \prod \lambda_n$ || eigenvalues of $X^TX \ge 0$.

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix M: $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T\mathbf{M}\mathbf{z} > 0\ [\ge 0]$ $\Leftrightarrow (> 0)$ [1] n positive eigenvalues. [2] full rank. [3] invertible

Score Func / Vector U	Single: $U(\theta,\mathbf{Y}) = \ell'(\theta;\mathbf{Y})$	$\mathbb{E}[U] = 0$	$Var[U] = \mathcal{I}(\theta)$
	Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1}\ell(\theta;\mathbf{Y}), ..., \partial_{\theta_p}\ell(\theta;\mathbf{Y})]^T \in \mathbb{R}^p$	$\mathbb{E}[\mathbf{U}] = \mathbf{0}$	$Var[\mathbf{U}] = \mathcal{I}(\theta)$ $I_{j,k} = -Cov(\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_j}, \frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_k}) = -\mathbb{E}[\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_j}, \frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_k}]$
CLT of U	Single: $U(\theta_0;\mathbf{Y}) \sim \mathcal{N}(0, \mathcal{I}(\theta_0))_{,n\to\infty}$		
	Multiple: $\mathbf{U}(\theta_0;\mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, \mathcal{I}(\theta_0))_{,n\to\infty}$	Also: $\mathbf{U}(\theta_0;\mathbf{Y})^T\mathcal{I}^{-1}(\theta_0)\mathbf{U}(\theta_0;\mathbf{Y}) \sim \chi^2_p$	

Fisher Information $\mathcal{I}$	Single: Observed: $J(\theta,\mathbf{y}) = -U'(\theta,\mathbf{y}) = -\ell''(\theta;\mathbf{y})$	Expected: $\mathcal{I}(\theta) = -\mathbb{E}[\ell''(\theta;\mathbf{Y})]$ || Unit: $i(\theta) = \mathcal{I}(\theta)/n = -\mathbb{E}[\ell''(\theta;Y_i)]$
	Multiple: Observed: $\mathbf{J}(\theta;\mathbf{y}) \in \mathbb{R}^{p\times p}, J_{j,k} = -\frac{\partial^2\ell(\theta,\mathbf{y})}{\partial\theta_j\,\partial\theta_k}$	Expected: $\mathcal{I}(\theta) \in \mathbb{R}^{p\times p}, \mathcal{I}_{j,k} = -\mathbb{E}[\frac{\partial^2\ell(\theta,\mathbf{Y})}{\partial\theta_j\,\partial\theta_k}]$ ps: $\mathcal{I}(\theta)$ always positive semi-definite
$\mathcal{I}\uparrow \Rightarrow \mathbf{Y}$ 对 θ 越敏感；信息量大；参数估计更精确		A model is $identifiable$ iff $\mathcal{I}(\theta)$ is positive definite. ps:(一维) identifiable if $\mathcal{I}(\theta) \ne 0$
CLT of MLE / CLT of I	Single: $\mathcal{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, \mathcal{I}^{-1}(\theta_0))_{,n\to\infty}$	$\widehat{Var}(\hat{\theta}) \approx \mathcal{I}^{-1}(\hat{\theta})$ | $ESE(\hat{\theta}) \approx \sqrt{\mathcal{I}^{-1}(\hat{\theta})}$ estimated standard error
	Multiple: $\mathcal{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim N_p(\theta_0, \mathcal{I}^{-1}(\theta_0))_{,n\to\infty}$	Also: $(\hat{\theta} - \theta_0)^T\mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) \sim \chi^2_p$

2.1 For Single Parameter

*__Random Variable / Function__: $L(\theta;\mathbf{Y})$ is random function (NOT pdf). For fixed θ, it's r.v. $\hat{\theta}(\mathbf{Y}) = ArgMax_{\theta\in\Theta}[L(\theta;\mathbf{Y})]$ is r.v. ps: $\mathcal{I}^{-1}, J^{-1}$ 是指其负一次方，不是反函数 True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0;\mathbf{Y}) \sim \mathcal{N}(0, \mathcal{I}(\theta_0))$ $T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{\mathcal{I}(\theta_0)}} \sim \mathcal{N}(0,1)$	$H_0: \theta = \theta_0$ 不用 MLE, 泰勒展开	$\left\{\theta : \frac{U^2(\theta)}{\mathcal{I}(\theta)} < \chi^2_{\alpha,1}\right\}$ or(approx):$\hat{\theta} \pm z_{\alpha/2}\sqrt{I^{-1}(\hat{\theta})}$	or $T^2_{Score}(\mathbf{Y}) \sim \chi^2_1$ $T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{I(\hat{\theta})}}, \frac{U(\theta_0)}{\sqrt{J(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{J(\hat{\theta})}}$
Wald Test or Maximum Likelihood(ML) Test	$T_{Wald}(\mathbf{Y}) = \sqrt{\mathcal{I}(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta}-\theta_0}{\sqrt{\mathcal{I}^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0,1)$	$H_0: \theta = \theta_0$ 构建 CI	$\{\theta : (\hat{\theta} - \theta)^2\mathcal{I}(\theta) < \chi^2_{\alpha,1}\}$ or(approx): $\hat{\theta} \pm z_{\alpha/2}\sqrt{I^{-1}(\hat{\theta})}$	or $T^2_{Wald} \sim \chi^2_1$; $_{n\to\infty}$ $T_{Wald} \simeq T_{Score}$ $T_{Ward}(\mathbf{Y}) \approx \frac{(\hat{\theta}-\theta_0)}{\sqrt{I^{-1}(\hat{\theta})}}, \frac{(\hat{\theta}-\theta_0)}{\sqrt{J^{-1}(\theta_0)}}, \frac{(\hat{\theta}-\theta_0)}{\sqrt{J^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi^2_1$ as $n \to \infty$ $LR = L(\theta_0)/L(\hat{\theta})$, $T_{LR} = -2ln(LR)$	$H_0: \theta = \theta_0$ 最小 CI, Most Powerful	$\left\{\theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < \chi^2_{\alpha,1}\right\}$	ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \simeq (\theta_0 - \hat{\theta})^2J(\hat{\theta})$
$T_{Score} \simeq T_{Wald} \simeq T_{LR}$ as $n \to \infty$; $J(\theta_0) \approx \mathcal{I}(\theta_0)$ as $n \to \infty$; When $\ell(\theta) =$ 关于 θ 的二次函数 $\Rightarrow$ All same. Geometric:(T_S, 比斜率),(T_W, 比宽度),(T_{LR}，比高度)				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{\mathcal{I}(\theta_r)}$
Advantages	Converge *very quickly* near optimum	$\mathcal{I}$ is constant / easier to calculate; Converge for wider θ_0
Stoping Rule	Given tolerance level ε: [1] Stop after r if $\|U(\theta_r)\| \le \varepsilon$ (Common) or [2] Stop after $\|\theta_r - \theta_{r-1}\| \le \varepsilon$	

2.3 For Multiple Parameters

* 此节中，θ 代表向量,$\theta = (\theta_1, ..., \theta_p)$ $\mathcal{I}^{-1}$ 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T^2_{Score}, T^2_{Wald}, T_{LR}$)	Hypothesis	(Confidence Region) CR
Score Test	$\mathbf{U}(\theta_0)^T\mathcal{I}^{-1}(\theta_0)\mathbf{U}(\theta_0) > z_{\alpha,p}$	$\theta = \theta_0$	$\{\theta : \mathbf{U}(\hat{\theta})^T\mathcal{I}^{-1}(\theta)\mathbf{U}(\hat{\theta}) < z_{\alpha,p}\}$
Wald Test	$(\hat{\theta} - \theta_0)^T\mathcal{I}(\theta_0)(\hat{\theta} - \theta_0) > z_{\alpha,p}$	$\theta = \theta_0$	$\{\theta : (\hat{\theta} - \theta)^T\mathcal{I}(\theta)(\hat{\theta} - \theta) < z_{\alpha,p}\}$
log-Likelihood Ratio Test	$2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha,p}$	$\theta = \theta_0$	$\{\theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha,p}\}$
$\mathcal{I}(\theta_0) \approx \mathcal{I}(\hat{\theta}), \mathbf{J}(\theta_0), \mathbf{J}(\hat{\theta})$ $ps: z_{\alpha,p} = \chi^2_{\alpha,p}$ T_{LR}:simple hypothesis testing.($dim(\mathcal{A}) = 0$) θ 都是向量			

Marginal Normal distribution $\hat{\theta}$:

$\hat{\theta}_j \sim \mathcal{N}(\theta_j, [\mathcal{I}^{-1}(\underline{\theta_0})]_{jj})$ θ_j: true para

$\cdot ESE(\hat{\theta}_j) = \sqrt{[\mathcal{I}^{-1}(\hat{\theta})]_{jj}}$

· CI(θ_j): $\hat{\theta}_j \pm z_\alpha ESE(\hat{\theta}_j)$

Composite Hypothesis Test: $H_0: \underline{\theta} \in \mathcal{A}, H_1: \underline{\theta} \notin \mathcal{A}$ $\mathcal{A}$: reduced model $LR = \frac{max_{\underline{\theta}\in\mathcal{A}}L(\theta)}{max_{\underline{\theta}\in\Theta}L(\theta)} = \frac{L(\hat{\underline{\theta}}_{\mathcal{A}})}{L(\hat{\underline{\theta}}_{\Theta})}$ ps: 多维未知参数,H_0 只想其中几个时用 || $dim(\mathcal{A})$ 常数时为 0

· **Generalised Likelihood Ratio Test**: $T_{GLR} = -2ln(LR) = 2\left(\ell(\hat{\underline{\theta}}) - \ell(\hat{\underline{\theta}}_{\mathcal{A}})\right) \sim \chi^2_{p-r}$ p : # of parameters. $r = dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0) *此节中，θ 代表向量	Newton-Raphson Method	Fisher's Method of Scoring
Formula	$\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r)\mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r)\mathbf{U}(\theta_r)$	$\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1}\mathbf{U}(\theta_r) = \theta_r + \mathcal{I}^{-1}(\theta_r)\mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$ || If $\mathcal{I} = diag(a, b, ...)$,$\Rightarrow \mathcal{I}^{-1} = diag(a^{-1}, b^{-1}, ...)$		

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider $\underline{\theta}$ as a r.v.. Bayesian approach ⇒ Probability are used to represent degree of believe about the unknown $\underline{\theta}$

Prior Distribution $p(\underline{\theta})$: Prior distribution $p(\underline{\theta})$ contains all available prior information about $\underline{\theta}$ *before observing any data*.

· **Hyper-Parameters**: 由于 $p(\underline{\theta})$ 是已知, 服从某种分布 (with parameter γ). We denoted $p(\underline{\theta})$ as $p(\underline{\theta};\gamma)$ where γ is *Hyper-parameters*. 定义: $\gamma_{prior} \to \gamma_{post}$

Likelihood function $p(\mathbf{y}|\underline{\theta}) = f(\mathbf{y}|\underline{\theta})$: Any observed data is considered as a realisation of r.v. Y with $f(\mathbf{y}|\underline{\theta})$

Posterior Distribution $p(\underline{\theta}|\mathbf{y})$: Posterior distribution $p(\underline{\theta}|\mathbf{y})$, contains all available knowledge.

· **Bayes' Rule / Theorem**: $p(\underline{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\underline{\theta})p(\underline{\theta})}{p(\mathbf{y})} = \frac{f(\mathbf{y}|\underline{\theta})p(\underline{\theta})}{m(\mathbf{y})}$ $\quad p(\mathbf{y}) = m(\mathbf{y}) = \int_{\underline{\theta}\in\Theta} f(\mathbf{y}|\underline{\theta})p(\underline{\theta})d\underline{\theta}$ (Normalizing constant)

Name	MAP Maximum A Posterior	MMSE Posterior Mean / Bayesian Minimum Mean Squared Error Estimator	Posterior Median $\widehat{\theta}_{median}$
Single Para	$\widehat{\theta}_{MAP} = \arg\max_\theta p(\theta\mid\mathbf{y})$	$\widehat{\theta}_{MMSE} = \mathbb{E}[\theta\mid\mathbf{y}] = \int_{\theta\in\Theta}\theta p(\theta\mid\mathbf{y})d\theta$ ps: 估算后验分布参数一般方法	$\frac{1}{2} = \int_{-\infty}^{\widehat{\theta}_{median}} p(\theta\mid\mathbf{y})d\theta = \int_{\widehat{\theta}_{median}}^{\infty} p(\theta\mid\mathbf{y})d\theta$

Name	Posterior Var	$C_\alpha = [a,b]$ $(1-\alpha)100\%$ Bayesian Credible Intervals (or Simply Credible Intervals)
Single Para	$Var[\theta\mid\mathbf{y}] = \int_{\theta\in\Theta}(\theta - \mathbb{E}[\theta\mid\mathbf{y}])^2 p(\theta\mid\mathbf{y})d\theta$	$\mathbb{P}(a \le \theta \le b\mid\mathbf{y}) = \int_a^b p(\theta\mid\mathbf{y})d\theta = 1-\alpha$ ps: $\mathbb{P}(\theta < a\mid\mathbf{y}) = \mathbb{P}(\theta > b\mid\mathbf{y}) = \alpha/2$

Bayesian Decision Theory: 目的: ^{1}Estimating parameter θ ^{2}Test Hypothesis about θ ^{3}Find CI. ps: 后两种有很多种方法来判断

1. Choose a *Loss Function* $Q(\widehat{\theta},\theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)
2. Minimises Posterior Expected Loss Function: $\widehat{\theta}_Q(\mathbf{y}) = \arg\min_{\widehat{\theta}} \mathbb{E}[Q(\widehat{\theta},\theta)|\mathbf{y}] = \arg\min_{\widehat{\theta}} \int_{\theta\in\Theta} Q(\widehat{\theta},\theta)p(\theta|\mathbf{y})d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For $\underline{\theta}$, $p(\underline{\theta})$ is *Conjugate Prior* for $p(\mathbf{y}|\underline{\theta}) = f(\mathbf{y}|\underline{\theta})$ iff 对 $p(\underline{\theta})$ 用 Bayes' Rule 得到的 $p(\mathbf{y}|\underline{\theta})$ 都在同一类型的分布

Conjugate Family: The collection of all *Conjugate Prior*.

Choice of Prior: Observed data number n: $^1 n$ small ⇒ Posterior shows a great dependency of the choice of prior.

$^2 n$ middle ⇒ posterior less affected by choice of prior. $^3 n$ large ⇒ choice of prior has limited effect on posterior.(Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: ^{1} $\widehat{\theta}_{MMSE} \approx \widehat{\theta}_{MAP}$ for most cases, especially n large. $^2 n$ large: $\widehat{\theta}_{MAP} \approx \widehat{\theta}_{MLE}$, $CI \approx C_\alpha$

3.3 Approximate Bayesian Inference

Intractable Posterior: Non-Conjugate model. (not analytic, no closed-form)

Monte Carlo Methods: Any func $g(\underline{\theta})$ || $I_g = \int_{\underline{\theta}\in\Theta} g(\underline{\theta})p(\underline{\theta}|\mathbf{y})d\underline{\theta} \approx \frac{1}{M}\sum_{i=1}^M g(\underline{\theta}^{(i)}) \xrightarrow{a.s.} I_g$ $\quad \underline{\theta}^{(i)} \sim p(\underline{\theta}|\mathbf{y}), i = 1,\dots,M$

1. **Importance Sampling (IS)**: ^{1} Simulating samples from another distribution $q(\underline{\theta})$ (*Proposal Distribution*) Consistent, Biased for finite

   $^2 w^{(i)} = \frac{\pi(\underline{\theta}^{(i)};\mathbf{y})}{q(\underline{\theta}^{(i)})}$, $\pi(\underline{\theta};\mathbf{y}) = f(\mathbf{y}|\underline{\theta})p(\underline{\theta})$ $\quad ^3$ IS estimator: $\hat{I}_g^M = \sum_{i=1}^M g(\underline{\theta}^{(i)})\overline{w}^{(i)} \xrightarrow{a.s.} I_g$, Normalized Weight: $\overline{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^M w^{(j)}}$ $\quad \underline{\theta}^{(i)} \sim q(\underline{\theta})$ ps: $\sum_{j=1}^M \overline{w}^{(j)} = 1$

   Remark: Choice of $q(\underline{\theta})$ is important. || $\hat{I}_g^M$ depends on: ^{1}# of M; ^{2}mismatch between proposal and the integrand of the integral. Good if $q(\underline{\theta}) \propto g(\underline{\theta})p(\underline{\theta}|\mathbf{y})$, difficult to achieve.:(
2. **Markov Chain Monte Carlo (MCMC)**: MCMC 算法通过 Markov Chain 生成样本, 使得样本接近服从后验分布.(计算量大)

Laplace Approximation: (最简单的估算 posterior 的方法) $p(\underline{\theta}|\mathbf{y}) \approx \mathcal{N}(\underline{\theta}_{MAP}; H^{-1}(\underline{\theta}_{MAP}))$ 类似 MLE 的 J $\quad H(\underline{\theta}) = -\frac{d^2 ln(p(\underline{\theta}|\mathbf{y}))}{d\underline{\theta}^2}$ (适合 $p(\underline{\theta}|\mathbf{y})$ 接近正态分布的时用)

Variational Inference (VI): ^{1}Choose $q(\underline{\theta};\underline{\eta})$, $\underline{\eta}$ 是 q 的参数. ^{2} 选择衡量 $p(\underline{\theta}|\mathbf{y}), q(\underline{\theta};\underline{\eta})$ 差异的方法 e.g.**KL divergence**: $KL(q(\underline{\theta};\underline{\eta})||p(\underline{\theta}|\mathbf{y})) = \int_{\theta\in\Theta} q(\underline{\theta};\underline{\eta})ln(\frac{q(\theta;\eta)}{p(\theta|\mathbf{y})})d\underline{\theta}$

^{3} 优化参数 (Find parameters to minimise the divergence): $\underline{\eta}^* = \arg\min_{\eta\in\mathcal{A}_\eta} KL(q(\underline{\theta};\underline{\eta})||p(\underline{\theta}|\mathbf{y}))$ $\quad ^4 p(\underline{\theta}|\mathbf{y}) \approx q(\underline{\theta};\underline{\eta}^*)$

Remark: Both Laplace Approximation & Variation Inference approximate $p(\underline{\theta}|\mathbf{y})$ by a parameter family of distributions. || 拉普拉斯是正态分布家族,VI 是更普遍的分布家族

4 Linear Model

4.1 Simple Linear Model ($y = \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \beta x_i$ $\quad i = 1,2,..,n$ || Suppose $^1\varepsilon_i$ independent $^2\mathbb{E}[\varepsilon_i] = 0$ $^3 Var[\varepsilon_i] = \sigma^2$ || Y_i response variable, x predictor variable.

Least Square (LS) Estimation	RSS Residual Sum of Squares: $S(\beta) = \sum(y_i - \mu_i)^2 = \sum(y_i - x_i\beta)^2$	LS Method: $\widehat{\beta}_i = \arg\min_\beta S(\beta)$ ps: 求偏导 =0 $\Vert$ $\widehat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\widehat{\beta}] = \beta$,unbiased $\Vert$ $Var[\widehat{\beta}] = \sigma^2_{\widehat{\beta}} = (\sum x_i^2)^{-1}\sigma^2$ (σ^2 known)	$\widehat{Var}[\widehat{\beta}] = \widehat{\sigma^2}_{\widehat{\beta}} = (\sum x_i^2)^{-1}\cdot\widehat{\sigma^2}$ $\Vert$ $\widehat{\sigma^2} = \frac{1}{n-1}\sum(y - x_i\beta)^2$
H_0, H_1 Test $\Vert$ CI	$T = (\widehat{\beta} - \beta_0)\,/\,Var(\widehat{\beta}) \sim \mathcal{N}(0,1)$ (σ^2 known)	$T = (\widehat{\beta} - \beta_0)\,/\,\widehat{Var}(\widehat{\beta}) \sim t_{n-1}$ (σ^2 unknown)
Other	**Fitted Values**: $\hat{\mu}_i = x_i\hat{\beta}$ $\Vert$ **Residuals**: $\widehat{\varepsilon}_i = y_i - \hat{\mu}_i$	For CI / PI / H_0/H_1 Test: Assume $\varepsilon_i \sim \mathcal{N}(0,\sigma^2)$
PI	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2}\sigma\sqrt{1 + \frac{x_0^2}{S_{xx}}}$	σ^2 unknown: $\hat{y}_{new} \pm t_{\alpha/2,n-1}\widehat{\sigma}_{\widehat{\beta}}\sqrt{1 + \frac{x_0^2}{S_{xx}}}$

4.2 Simple Linear Model ($y = \alpha + \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \alpha + \beta x_i$ $\quad i = 1,2,..,n$ || Suppose $^1\varepsilon_i$ independent $^2\mathbb{E}[\varepsilon_i] = 0$ $^3 Var[\varepsilon_i] = \sigma^2$

SL Estimate	$S_{xy} = \frac{1}{n}\sum(x_i - \overline{x})(y_i - \overline{y}) = \sum x_i y_i - \frac{1}{n}(\sum x_i)(\sum y_i)$	LS Method: $\widehat{\beta} = \frac{S_{xy}}{S_{xx}}$ $\vert$ $\widehat{\alpha} = \overline{y} - \widehat{\beta}\overline{x}$	$\widehat{\alpha}, \widehat{\beta} = \arg\min_{\alpha,\beta} S(\alpha,\beta)$ $\vert$ $S(\alpha,\beta) = \sum(y_i - \mu_i)$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\widehat{\beta}] = \beta$,unbiased $\Vert$ $Var[\widehat{\beta}] = \sigma^2_{\widehat{\beta}} = \frac{\sigma^2}{S_{xx}}$	$\widehat{Var}[\widehat{\beta}] = \widehat{\sigma^2}_{\widehat{\beta}} = \frac{\widehat{\sigma^2}}{S_{xx}}$	$\widehat{\sigma^2} = \frac{1}{n-2}\sum(y - \alpha - \beta x)^2 = \frac{1}{n-2}(S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
$\mathbb{E}, Var$ For α	$\mathbb{E}[\widehat{\alpha}] = \alpha$,unbiased $\Vert$ $Var[\widehat{\alpha}] = \sigma^2_{\widehat{\alpha}} = \sigma^2\left(\frac{1}{n} + \frac{\overline{x}^2}{S_{xx}}\right)$	$\widehat{Var}[\widehat{\alpha}] = \widehat{\sigma^2}_{\widehat{\alpha}} = \widehat{\sigma^2}\left(\frac{1}{n} + \frac{\overline{x}^2}{S_{xx}}\right)$	$\widehat{\sigma^2} = \frac{1}{n-2}\sum(y - \alpha - \beta x)^2 = \frac{1}{n-2}(S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
H_0, H_1 Test $\Vert$ CI	$T = (\widehat{\beta} - \beta_0)\,/\,Var(\widehat{\beta}) \sim \mathcal{N}(0,1)$ (σ^2 known)	$T = (\widehat{\beta} - \beta_0)\,/\,\widehat{Var}(\widehat{\beta}) \sim t_{n-2}$ (σ^2 unknown)	$T = (\widehat{\alpha} - \alpha_0)\,/\,\widehat{Var}(\widehat{\alpha}) \sim t_{n-2}$ (σ^2 unknown)
(ctd.) Other	**Fitted Values**: $\hat{\mu}_i = \hat{\alpha} + \hat{\beta}x_i$ $\Vert$ 残差: $\widehat{\varepsilon_i} = y_i - \hat{\mu}_i$ $\Vert$ $\sum\hat{\varepsilon}_i = 0, \sum\hat{\varepsilon}_i x_i = 0$	For CI / PI / H_0/H_1 Assume $\varepsilon_i \sim \mathcal{N}(0,\sigma^2)$	$T = (\widehat{\alpha} - \alpha_0)/Var(\widehat{\alpha}) \sim \mathcal{N}(0,1)$ (σ^2 known)
PI $\vert$ Cor.r = $\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2}\sigma\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{x})^2}{S_{xx}}}$	σ^2 已知: $\hat{y}_{new} \pm t_{\alpha/2,n-2}\sigma\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{x})^2}{S_{xx}}}$	σ^2 未知: $\hat{y}_{new} \pm t_{\alpha/2,n-2}\widehat{\sigma}_{\hat{y}}\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{x})^2}{S_{xx}}}$

ps: $Cov(\widehat{\alpha},\widehat{\beta}) = -\frac{\sigma^2\overline{x}}{S_{xx}}$ or $\widehat{\sigma}^2$ $Cov(\widehat{\alpha},\widehat{\beta}) = 0$ ⇒ independent. || $S_{yy} = SS_{regression} + SS_{residuals}$ where $SS_{regression} = \sum(\hat{y}_i - \overline{y})^2$, $SS_{residuals} = \sum(y_i - \hat{y}_i)^2$

ps: Correlation: $Cor(\widehat{\alpha},\widehat{\beta}) = \frac{Cov(\widehat{\alpha},\widehat{\beta})}{\sqrt{Var(\widehat{\alpha})Var(\widehat{\beta})}} = -\frac{\overline{x}}{\sqrt{\frac{1}{n}S_{xx} + \overline{x}^2}}$ or 使用估计的 ⇒ If = 0 ⇒ independent.

4.3 General Linear Model

Lemma|$\mathbb{E}$, Var in Matrix: If r.v. $\mathbf{v}$, $\mathbf{A}, \mathbf{b}$ constant $\Rightarrow$ $^1\mathbb{E}[\mathbf{Av}+\mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{v}]+\mathbf{b}$ $\quad ^2 Var[\mathbf{Av}+\mathbf{b}] = \mathbf{A}Var[\mathbf{v}]\mathbf{A}^T$ || ↓注意: 对于 fitted residual 同方差不成立;$\underline{\hat{\beta}}$ 和拟合残差独立

Metric Predictor: 设计矩阵用连续观测值. || **Factor Predictor**: 离散数据用指示函数 (0/1) 填满, 每个" 小组" 对应一个指示函数.

Linear Model: $Y_i = \mu_i + \varepsilon_i$, $i = 1, 2, .., n$ $\quad \mu = f(\underline{\beta}, \mathbf{x})$ || $f(\underline{\beta}, \mathbf{x})$ linear. || $\varepsilon_i \overset{i.i.d.}{\sim}$ distribution. $\mathbb{E}[\varepsilon_i] = 0$ $\quad Var[\varepsilon_i] = \sigma^2$

· **Matrix Expression**: $\underline{\mathbf{y}} = \mathbf{X}\underline{\beta} + \underline{\varepsilon}$ || Response Vector: $\underline{\mathbf{y}}$ | Design / Model Matrix: $\mathbf{X}$ | Parameters / Coefficients: $\underline{\beta}$ | Additive Noise Vector: $\underline{\varepsilon}$

· **Parameter Estimates**: $\underline{\hat{\beta}}$ | **Estimated / Fitted Value Mean Vector**: $\underline{\hat{\mu}} = \mathbf{X}\underline{\hat{\beta}}$ | **Estimated / Fitted Residual** Vector: $\underline{\hat{\varepsilon}} = \underline{\mathbf{y}} - \underline{\hat{\mu}}$

4.4 Linear Model Theory

Orthonormal Matrix $\mathcal{Q}$	Def: $\mathcal{Q}$:= columns and rows are orthonormal vectors.	$\mathcal{Q}^T = \mathcal{Q}^{-1}$	$\mathcal{Q}\mathcal{Q}^T = \mathcal{Q}^T\mathcal{Q} = I_n$	$\forall \mathbf{v}, \|\|\mathbf{v}\|\|^2 = \|\|\mathcal{Q}\mathbf{v}\|\|^2 = \|\|\mathcal{Q}^T\mathbf{v}\|\|^2$
(ctd.) ★★	Matrix X Orthogonal iff X^TX is diagonal matrix.	X^TX always symmetric matrix $\Rightarrow$ $\mathbf{V}_{\underline{\hat{\beta}}}$ is diagonal $\Rightarrow$ β_i independent		
QR Decomposition	$\forall\, \mathbf{X} \in \mathbb{R}^{n\times p} \Rightarrow \mathbf{X} = \mathcal{Q}\begin{pmatrix}\mathbf{R}\\0\end{pmatrix} = \mathbf{QR}$	$\mathbf{R} \in \mathbb{R}^{p\times p}$ is *upper triangular matrix*; $\mathcal{Q} \in \mathbb{R}^{n\times n}$ is *orthonormal matrix*; $\mathcal{Q}$ 前 p 列等于 $\mathbf{Q} \in \mathbb{R}^{n\times p}$		

Least Square (LS)	$RSS := S(\underline{\beta}) = \sum_{i=1}^n \varepsilon_i^2(\underline{\beta}) = \sum_{i=0}^n (y_i - \mu_i(\underline{\beta}))^2 = \|\|\mathbf{y} - \underline{\mu}\|\|^2 = \|\|\mathbf{y} - \mathbf{X}\underline{\beta}\|\|^2 = \|\|\underline{\varepsilon}(\underline{\beta})\|\|^2$	$\underline{\hat{\beta}} = \arg\min_\beta S(\beta) = \arg\min_\beta \|\|\mathbf{f} - \mathbf{R}\underline{\beta}\|\|^2$
QR Method of LS	Let $^1\mathcal{Q}$ be orthonormal matrix $\quad ^2\mathbf{X} = \mathbf{QR}$ $\quad ^3\mathcal{Q}^T\mathbf{y} = \begin{pmatrix}\mathbf{f} \in \mathbb{R}^p\\ \mathbf{r} \in \mathbb{R}^{n-p}\end{pmatrix}$ $\Rightarrow$ $S(\underline{\beta}) = \|\|\mathcal{Q}^T\underline{\varepsilon}(\underline{\beta})\|\|^2 = \cdots = \|\|\mathbf{f} - \mathbf{R}\underline{\beta}\|\|^2 + \|\|\mathbf{r}\|\|^2$	
QR Method (ctd.)	If $\mathbf{R}$ 满秩: $\underline{\hat{\beta}} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$ $\quad RSS = \|\|\mathbf{r}\|\|^2 = \|\|\underline{\varepsilon}(\underline{\hat{\beta}})\|\|^2 = \|\|\mathbf{y} - \mathbf{X}\underline{\hat{\beta}}\|\|^2$ $\quad RSE = \frac{RSS}{n-p} = \hat{\sigma}^2$	ps:$\mathbf{f} = \mathbf{Q}^T\mathbf{y}$ $\quad$ || $\mathbf{Q}^T\mathbf{Q} = I_p, \mathbf{QQ}^T \neq I$

More Inference of LS:

Matrix Method of LS	$S(\underline{\beta}) = (\mathbf{y} - \mathbf{X}\underline{\beta})^T(\mathbf{y} - \mathbf{X}\underline{\beta}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\underline{\beta} + \underline{\beta}^T\mathbf{X}^T\mathbf{X}\underline{\beta}$	$\underline{\hat{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	$\mathbf{V}_{\underline{\hat{\beta}}} = Var[\underline{\hat{\beta}}] = (\mathbf{X}^T\mathbf{X})^{-1}\sigma^2$

Influence / Hat Matrix A	$\mathbf{A} = \mathbf{QQ}^T = \mathbf{A}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$	$\mathbf{A}$ is symmetric. || # of (identifiable) parameters = $tr(\mathbf{A}) = tr(\mathbf{QQ}^T) = tr(I_p) = p$		
$Var[\mathbf{y}]$, $Var[\mathbf{f}]$, $\underline{\hat{\mu}}$, $\underline{\hat{\beta}}$	$Var[\mathbf{y}] = Cov(\underline{\varepsilon}, \underline{\varepsilon}) = \sigma^2 I_n$	$Var[\mathbf{f}] = \mathbf{Q}^T\mathbf{Q}\sigma^2 = I_p$	$\underline{\hat{\mu}} = \mathbf{X}\underline{\hat{\beta}} = \mathbf{Ay}$	$\underline{\hat{\beta}} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$
$\mathbf{V}_{\underline{\hat{\beta}}}$, $\mathbf{V}_{\underline{\hat{\varepsilon}}}$, $\mathbf{V}_{\underline{\hat{\mu}}}$ || $\underline{\hat{\mu}}$ dist	$\mathbf{V}_{\underline{\hat{\beta}}} = Var[\underline{\hat{\beta}}] = \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2$	$\mathbf{V}_{\underline{\hat{\varepsilon}}} = Var[\underline{\hat{\varepsilon}}] = (I_n - \mathbf{A})\sigma^2$	$\mathbf{V}_{\underline{\hat{\mu}}} = Var[\underline{\hat{\mu}}] = \mathbf{A}\sigma^2$	$\underline{\hat{\mu}}$ is Degenerate Normal (退化正太分布,$\mathbf{\Sigma}$ 不满秩)
$\mathbb{E}[\underline{\hat{\beta}}]$, $\mathbb{E}[\underline{\hat{\varepsilon}}]$, $\mathbb{E}[\underline{\hat{\mu}}]$ || $\underline{\hat{\varepsilon}}$ dist	$\mathbb{E}[\underline{\hat{\beta}}] = \underline{\beta}$ unbiased	$\mathbb{E}[\underline{\hat{\varepsilon}}] = 0$	$\mathbb{E}[\underline{\hat{\mu}}] = \underline{\mu}$ (unbiased)	$\underline{\hat{\varepsilon}}$ is Degenerate Normal (退化正太分布,$\mathbf{\Sigma}$ 不满秩)
检查正交/线性无关: If $\mathbf{X}^T\mathbf{X}$ is diagonal $\Rightarrow$ $\mathbf{X}$ Orthogonal. $\Rightarrow$ $\mathbf{V}_{\underline{\hat{\beta}}}$ diagonal $\Rightarrow$ β_i independent.			**Check Identifiable**: 法 1:$det(X^TX) \neq 0$,identifiable. 法 2:$Rank(X) = p$ 既列向量 independent.)	

Prediction | PI: ★★ Create a Prediction Matrix: $\mathbf{X}^*$ ($\mathbf{X}^*$ 形式和 $\mathbf{X}$ 保持一直, 数据变为按拟合后的每个点的数据) $\Rightarrow$ σ^2 已知 $\underline{\hat{\mu}}^* = \mathbf{X}^*\underline{\hat{\beta}}$, $\underline{\hat{\mu}}^* \sim \mathcal{N}(\underline{\mu}^*, X^*(X^TX)^{-1}(X^*)^T\sigma^2)$

4.5 CI / Hypothesis Testing

Normality Assumption: Assume that $\underline{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n)$ $\Rightarrow$ $^1\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\underline{\beta}, I_n\sigma^2)$ $\quad ^2\underline{\hat{\beta}} \sim \mathcal{N}_p(\underline{\beta}, \mathbf{V}_{\underline{\hat{\beta}}})$ || 对于 **Normal, LS $\underline{\hat{\beta}}$ = MLE $\underline{\hat{\beta}}$**

$\mathbf{V}_{\mathcal{Q}^T\mathbf{y}}$, $\mathbb{E}[\mathcal{Q}^T\mathbf{y}]$ || $\hat{\sigma}^2$, $\hat{\mathbf{V}}_{\underline{\hat{\beta}}}$	$\mathbf{V}_{\mathcal{Q}^T\mathbf{y}} = Var[\mathcal{Q}^T\mathbf{y}] = I_n\sigma^2$	$\mathbb{E}[\mathcal{Q}^T\mathbf{y}] = \mathbb{E}\left[\begin{pmatrix}\mathbf{f}\\ \mathbf{r}\end{pmatrix}\right] = \begin{pmatrix}\mathbf{R}\\ \mathbf{0}\end{pmatrix}\underline{\beta}$	$\hat{\sigma}^2 = \frac{\|\|\mathbf{r}\|\|^2}{n-p}$ (unbiased)	$\hat{\mathbf{V}}_{\underline{\hat{\beta}}} = \mathbf{R}^{-1}\mathbf{R}^{-T}\hat{\sigma}^2$ (unbiased) || $\underline{\hat{\beta}}, \hat{\sigma}^2$ independent
$\mathbf{f}, \mathbf{r}$ Distribution	Known σ^2: $\mathbf{f} \sim \mathcal{N}(\mathbf{R}\underline{\beta}, I_p\sigma^2)$ $\quad \mathbf{r} \sim \mathcal{N}(\mathbf{0}, I_{n-p}\sigma^2)$		Unknown σ^2: $\hat{\sigma}_{\hat{\beta}_i} = \sigma_{\hat{\beta}_i}\hat{\sigma}/\sigma$ || $\hat{\sigma}_{\hat{\beta}_i} = \sqrt{[\hat{\mathbf{V}}_{\underline{\hat{\beta}}}]_{ii}}$ $\quad \sigma_{\hat{\beta}_i} = \sqrt{[\mathbf{V}_{\underline{\hat{\beta}}}]_{ii}}$	
Test Statistic	For $\underline{\beta}$ 整体: $T_{known\ \sigma^2}(\mathbf{Y}) = \frac{1}{\sigma^2}\|\|\mathbf{r}\|\|^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n-p} r_i^2 \sim \chi^2_{n-p}$		For β_i 个体: $T_{unknown\ \sigma^2}(\mathbf{Y}) = (\hat{\beta}_i - \beta_i)/\hat{\sigma}_{\hat{\beta}_i} \sim t_{n-p}$	

Testing Constraints in the Model: 如果我们只是想对一部分 β_i 进行假设检验 H_0, H_1. 使用此方法.

[1] Create Selection Matrix $\mathbf{C} \in \mathbb{R}^{q\times p}$ q 检验个数,p 参数总数

e.g. $\mathbf{C} = \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{pmatrix}$ 如果我们要检验 β_1, β_2 但总数有 3 个参数.

[2] Create $\mathbf{d}$ s.t. $\mathbf{C}\underline{\beta} = \mathbf{d}$ $\quad$ $\mathbf{d}$ 即我们希望检验 β_i 的数值.

[3] Under H_0: $\mathbf{C}\underline{\hat{\beta}} - \mathbf{d} \sim \mathcal{N}(\mathbf{0}, \mathbf{CV}_{\underline{\hat{\beta}}}\mathbf{C}^T)$ || $H_0 : \mathbf{C}\underline{\beta} = \mathbf{d}$ vs $H_1 : \mathbf{C}\underline{\beta} \neq \mathbf{d}$

$\mathbf{CV}_{\underline{\hat{\beta}}}\mathbf{C}^T = \mathbf{L}^T\mathbf{L}$ (Cholesky Decomposition)

$\mathbf{L}^{-T}(\mathbf{C}\underline{\hat{\beta}} - \mathbf{d}) \sim \mathcal{N}(\mathbf{0}, I_q)$

★ $RSS_{reduced} = S(\underline{\hat{\beta}}_{reduced})$ $\quad RSS_{full} = S(\underline{\hat{\beta}}_{full})$

[4] Test Statistic: $\quad$ ps:$T > F_\alpha$ 拒绝

(σ^2 known): $T(\mathbf{Y}) = (\mathbf{C}\underline{\hat{\beta}} - \mathbf{d})^T(\mathbf{CV}_{\underline{\hat{\beta}}}\mathbf{C}^T)^{-1}(\mathbf{C}\underline{\hat{\beta}} - \mathbf{d}) \sim \chi^2_q$

(σ^2 unknown): $T(\mathbf{Y}) = \frac{1}{q}(\mathbf{C}\underline{\hat{\beta}} - \mathbf{d})^T(\mathbf{C}\hat{\mathbf{V}}_{\underline{\hat{\beta}}}\mathbf{C}^T)^{-1}(\mathbf{C}\underline{\hat{\beta}} - \mathbf{d}) \sim F_{q,n-p}$

(if $\mathbf{d} = \mathbf{0}$) $= \frac{(RSS_{reduced} - RSS_{full})/q}{RSS_{full}/(n-p)} \sim F_{q,n-p}$

4.6 Assumptions || Diagnostics

R^2 , R^2_{adj} **Metric**: $R^2 = 1 - \frac{\frac{1}{n}RSS}{\frac{1}{n}\sum_{i=1}^n(y_i - \bar{y})^2} = 1 - \frac{\frac{1}{n}\sum_{i=1}^n \hat{\varepsilon}_i^2}{\frac{1}{n}\sum_{i=1}^n(y_i - \bar{y})^2}$ $\quad RSS = S(\underline{\hat{\beta}}_{full})$ (接近 1, 拟合效果好) || $R^2_{adj} = 1 - \frac{\frac{1}{n-p}\sum_{i=1}^n \hat{\varepsilon}_i^2}{\frac{1}{n-1}\sum_{i=1}^n(y_i - \bar{y})^2}$ (avoid biased of R^2)

Graphical Assessment

1. Residuals vs Fitted (Check linear assumption): $(\hat{\mu}_i, \hat{\varepsilon}_i)$

   [1] 平均值线在 0 附近, 越靠 0 越好. ($\Rightarrow \mathbb{E}[\hat{\varepsilon}_i] = 0$) || [2] 有趋势/系统性 (systematic pattern in the mean of the residuals): violate the independence assumption.

   [3] 点在图上均匀分布 (spread / variability does not change much for different fitted values)→constant variance
2. Scale-Location (Check constant variance assumption): $(\hat{y}_i, \sqrt{|e_i^*|})$, $e_i^* = \frac{e_i}{s_e}$ $\quad$ 越靠近 0.82 越好.(around a constant value)
3. Normal QQ-Plot (Check Normal Assumption): (z_i, e_i^*) , $z_i = \Phi^{-1}(\frac{i-0.5}{n})$
4. Residuals vs Leverage (Check influential points): $(\mathbf{A}_{ii}, \hat{\varepsilon}_i/(\hat{\sigma}\sqrt{1 - \mathbf{A}_{ii}}))$ high leverage points: x 轴上远离大多数数据点.(对拟合结果影响大) $\quad$ 超过 cook's distance 认为影响大.
5. 线性回归图像 (Check Independence assumption): $(\mathbf{x}, \mu_i)$ or (x_i, μ_i) $\quad$ (符合假设 → 随机在回归线上随机分布的; 如果存在聚集性点 (cluster points)⇒ 不符合 independence.)

Check balance of quantiles: 检查残差的分布是否对称. Method 1: $\frac{|\hat{\varepsilon}_{median}|}{\hat{\sigma}}$ $\quad$ Method 2: $\frac{|\hat{\varepsilon}_{Q3}| - |\hat{\varepsilon}_{Q1}|}{\hat{\sigma}}$ small ⇒ no evidence against residuals not being symmetric

5 Distribution Table

Distribution	PDF	Mean	Var	Other
Bernoulli $X \sim Bern(p)$	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n, p)$
Binomial $X \sim Bin(n, p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$X, Y \sim Bin(n, p), Bin(m, p)$ $X + Y \sim Bin(n + m, p)$
For $X \sim Bin(n, p), n \to \infty, p \to 0, np \to \lambda,$ then $X \sim Poission(\lambda)$			For $X \sim Bin(n, p), n \to \infty$ and pdf symmetric, $\Rightarrow X \sim N(np, np(1-p))$	
Geometric $X \sim Geom(p)$	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	$\mathbb{E}[X^2] = 2/p^2$
Negative Binomial $X \sim NB(r, p)$	$\binom{r+x-1}{x}(1-p)^r p^x$, $r \geq 1$	$\frac{rp}{1-p}$	$\frac{rp}{(1-p)^2}$	或者 $p = 1 - q$, 注意!
Poisson $X \sim Po(\lambda)$	$e^{-\lambda} \cdot \frac{\lambda^x}{x!}$	λ	λ	$\mathbb{E}[X^2] = \lambda^2 + \lambda$
For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X + Y \sim Po(\lambda_1 + \lambda_1)$				

Multinomial $x_1+\cdots+x_k=n$ $(X_1,\ldots,X_k) \sim MN(n,p_1,\ldots,p_k)$	$\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$, $0<p_i<1$	np_i	$np_i(1-p_i)$	
Uniform $X \sim \mathcal{U}([a,b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[X^2] = \frac{b^2+ab+a^2}{3}$
Normal $X \sim \mathcal{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbf{Y} \sim N_k(\mu,\Sigma)$, μ vector $f_{\mathbf{Y}}(y_1,\ldots,y_k) = \frac{e^{-\frac{1}{2}(\mathbf{y}-\mu)^T\Sigma^{-1}(\mathbf{y}-\mu)}}{\sqrt{(2\pi)^k det(\Sigma)}}$
For $X,Y \sim N(\mu_1,\sigma_1^2), Po(\mu_2,\sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}$
Beta $X \sim Beta(\alpha,\beta), \alpha,\beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, x \in [0,1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $Beta(\alpha,\beta)$ has maximum probability, $x = (\alpha-1)/(\alpha+\beta-2)$				
Gamma $X \sim \Gamma(\alpha,\beta), \alpha,\beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha-1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2},\frac{1}{2})}(1+\frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0,1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution) When $n \to \infty$ it will converges to $N(0,1)$. **Cauchy Distribution**: $X \sim t_1$ and its mean and var are undefined.				
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}}\Gamma(\frac{v}{2})}x^{\frac{v}{2}-1}e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0,1)$, $then \sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$, $then\ X+Y \sim \chi_{n+m}^2$				
F $X \sim F(\lambda,v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}}v^{\frac{v}{2}}}{B(\frac{\lambda}{2},\frac{v}{2})}x^{\frac{\lambda}{2}-1}(\lambda x+v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1,Y_2 \sim \chi_{n_1}^2,\chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1,n_2}, X^{-1} \sim F_{n_2,n_1}$

6 Review

6.1 Basic Concepts

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error
Def	$\mathbb{P}(Rej\ H_0 \mid H_0\ is\ T)$	$\mathbb{P}(Fail\ Rej\ H_0 \mid H_0\ is\ F)$

-	H_0 is T	H_0 is F
Fail to Rej H_0	Right	Type II (β) 与 size 有关
Rej H_0	Type I (α) 与 size 无关	Right (Power)

Significance Level: $\alpha = \mathbb{P}(Rej\ H_0|H_0\ is\ T)$

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2\min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test: $\mathbb{P}$(Rej $H_0|H_0$ is F)$= 1-\beta = \mathbb{P}(Z \in C|\mu = \mu^*)$ · 1.Sample size ↑, Power ↑; 2.σ^2 ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4.α↑, Power ↑.

Def of Confidence Interval (CI): $100(1-\alpha)\%$ Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1-\alpha$.

(Convergence of r.v.) Name	Definition	written as	Example	
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \overset{d}{\to} X$ or $X_n \overset{d}{\to} F_X$	CLT 最弱:	见下面
Convergence in Distribution	If $\forall\epsilon > 0, \lim \mathbb{P}(\lvert X_n - X\rvert > \epsilon) = 0$.	$X_n \overset{p}{\to} X$	WLLN 中等:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$. $\Rightarrow \overline{X}_n \overset{p}{\to} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \overset{a.s.}{\to} X$	SLLN 最强:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$, $Var[X_i] < \infty. \Rightarrow \overline{X}_n \overset{a.s.}{\to} \mu$

Central Limit Theorem (CLT): If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty$. $\Rightarrow \frac{\overline{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \overset{d}{\sim} N(0,1)$ 样本均值的分布会趋近于正态分布

6.2 Test Statistics and Decision Rule

One Sample		Two Sample		
Find μ		Find μ		Find σ^2
Z-Test (σ^2 known) $Z = \frac{\bar{X}-\mu_0}{\sqrt{\sigma^2/n}} \sim N(0,1)$	t-Test (σ^2 unknown) $T = \frac{\bar{X}-\mu_0}{\sqrt{S^2/n}} \sim t_{n-1}$	Paired t-Test (X,Y same Unit) $D_i = Y_i - X_i \sim N(\mu_D, \sigma_D^2)$ $T = \frac{\bar{D}-\mu_D}{\sqrt{S^2/n}} \sim t_{n-1}$	Two-Sample t-Test (独立) $T = \frac{(\bar{X}-\bar{Y})-\Delta_0}{\sqrt{\sigma^2(\frac{1}{m}+\frac{1}{n})}} \sim N(0,1)$ 未知:S_p^2;$T \sim t_{m+n-2}$	F-Test $F = \frac{S_X^2}{S_Y^2}$ $\sim F_{m-1,n-1}$

· 对于 Two-Sample t-Test 我们要求 X,Y σ^2 一致 ps:$H_0 : \mu_D = \mu_X - \mu_Y$; $H_0 : \Delta_0 = \mu_X - \mu_Y$; $X : m, Y : n$

Pooled Sample Variance: (Two sample Variance) $S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}$ ps:Unbiased

7 Writing Format

Distribution|Unbiased|Consistent

寻找新 distribution: 产生新 distribution 的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}[(A\cap B)/B]$

Def|Unbiased:An estimator is unbiased if, on average, it equals the true value of the parameter it's estimating.

Def|Consistent: An estimator is consistent if it converges in probability to the true value of the parameter p as the sample size increases indefinitely.

MLE

Assumption|Calculate Estimator: Let random variables sequence $X_1, ..., X_n$ represent independent $\{?: e.g. sample\}$ from a common distribution {?}, represented by random variable X, that depends on the population parameter {?}.

Assumption|Calculate Maximum Likelihood: Suppose that we have n independent observations $x_1, ..., x_n$ drawn from an {?} distribution, with probability density function $f_x = \{?\}$.

Reason|$\hat{\sigma} = \sqrt{\hat{\sigma^2}}$: It's due to the invariant property of MLEs.

Reason|Newton's Method = (≠) Scoring Method: Because the second derivative of likelihood function doesn't (does) include the r.v.

Reason|$\mathbb{P}(\theta < ?) = \alpha$ is false: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).

Geometric Comparison of T_{Score},T_{Wald},T_{LR}:

· **1**.T_{Score} (Slope Comparison): Represents the slope of the likelihood function at θ_0; A steeper slope indicates that the likelihood function is changing more rapidly at $theta_0$, which implies a greater difference between $\widehat{\theta}$ and θ_0.

· **2**.T_{Wald} (Width Comparison): Compares the difference between $\widehat{\theta}$ and θ_0.

· **3**.T_{LR} (Height Comparison): Compares the MLE $L(\widehat{\theta})$ and the likelihood value at $\theta_0, L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\widehat{\theta}$ and θ_0

Reason|T_{Score},T_{Wald},T_{LR} Other 部分公式可以近似的理由: This is due to $\widehat{\theta}$ being a consistent estimator of θ_0, and $J(\theta_0)$ being a consistent estimator of $I(\theta_0)$

CI|PI

Def|CI: An interval of uncertainty for fitted regression line, given the explanatory variable x_0.

Meaning|CI 表示含义: Not Reject: If data are generated under H_0, there is a probability $(1-\alpha)$ for the true θ_0 to be within the $(1-\alpha)100\%$ CI. (i.e. If we were to repeat out experiment 100 times, we would expect the true θ_0 to be within the derived confidence interval (not necessarily CI:(a,b)) {e.g. 95} times.)

Assumption|CI: Let $X_1, ..., X_n$ be independent and identically distributed $\{?: distribution\}$ random variables, where {?} is unknown/known ...

Reason| 误差来源: The source of uncertainty in (or in CI) an estimate comes from the principal of repeated sampling from the population.

Reason| 无误差:The true {?} population parameter is assumed to be fixed (not random)

Def|PI: An interval of uncertainty for a future observation y_0, given explanatory variable x_0.

Reason|CI < PI: PI > CI, because it's accounts for the additional variability of the single observations, since $\sigma^2 > 0$

Reason|$\bar{x} = (left + right)/2$: The confidence interval for the $\{\mu\}$ is centered on $\bar{x}$

Hypothesis|P-value

p-value: p-value measures the plausibility that the data were generated under H_0. **Low p-value**: low p-value suggests H_0 is incompatible with the data and hence likely to be false.

Hypothesis Testing: **Steps**: [1] State H_0, H_1; [2] Select Test Statistic T; [3] distribution of T under H_0; [4] observed test statistic; [5] critical value / region, p-value, or CI; [6] conclusion. || **Reject** H_0: We would reject H_0 in favour of the H_1. (So that ...) || **Fail to reject** H_0: We fail to reject H_0. Hence, there is strong evidence against the null hypothesis.

Bayes Inference

Meaning|Important Sampling (IS): It's a Monte Carlo technique that can be used when simulating from the posterior is not possible.

Meaning|Variation Methods: It minimizes a divergence between the posterior and the variation approximation.

Linear Regression

Explain|⋆Why considered to be linear: Since the responses are independent, have the same variance and the expected value of the response given covariates is linear in the unknown coefficients $\{?: e.g. \beta_0, \beta_1, \beta_2, ...\}$

Reason| 不说一个增加导致另一个...: Correlation is not Causation. A linear regression model can only describes correlated relationship.

Reason|MLE 和 LS 公式一样: The functional form of the (log-)likelihood being a sum of squared differences.

Assumption|Normal Linear Regression: Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.

Meaning|LS 数值大小的表示含义: $S(\beta)$ small: μ_i relatively close to the y_i; Large: μ_i far from their corresponding y_i.

Meaning|ε: The random component of the model is intended to capture the fact that if we gathered a replicate set of data, the β_i (coefficient) would not change, but the Y_i would, as a result of different measurement errors.

Reason| 使用 SL 来算 $\underline{\beta}$: Usually, Linear Model can be a system of n equations with p parameters with $n > p$. In this case, the system is overdetermined and there is no exact solution. The least square method is used to find the best solution.

Geometric Interpretation of Linear Model|LS Method: The least Square Estimation of Linear Model amounts to finding the *orthogonal projection* of the n dimensional response data **y** onto the p dimensional linear subspace spanned by the columns of **X**.

Geometric Interpretation of Linear Model|$\mathbb{E}$[y]: $\mathbb{E}$[**y**] lies in the space spanned by the columns of **X**, | and the least square estimate seeks to find the point in this space that is closest to **y**.

Independent of $\underline{\widehat{\beta}}$, $\widehat{\sigma}^2$: f , r independent ⇒ $\underline{\widehat{\beta}}$, $\widehat{\sigma}^2$ independent.

When $Var(\varepsilon) = I_n\sigma^2$ (A = 0): Implies that ε_i independent.

Reason|R^2, R^2_{adj}: **1.** The linear model explains $\{R^2\}$ of the Variability within the $\{y\}$ data. **2.** $\{1-R\}$ of Variability of the $\{y\}$ data remains unexplained.

Reason|R^2, R^2_{adj}: R 越大 better predicts the data. || R^2 比较: As R^2 for the 'Model 1' is larger (smaller) than 'Model 2', so 'Model 1/2' is preferred as it explains more of the response variability.

Reason| 对于 Factor 类型的单个 p-value 不关心: If it's a model of Factors, it's often difficult to interpret the p-value for individual coefficients in any meaningful way. ⇒ Prefer to test the whole factor effect.

Reliability of the prediction: 计算 $k = \frac{|x_0 - \bar{x}|}{S_x}$ ps: 一般来说小于 1.96 以上才说好的 reliability.

· Linear Regression 从 $\bar{x}$ 往左右,reliable 逐渐下降.

Reason|y 和 x 之间是否有任何关联: Association between the explanatory variable and the response is measured by the F statistic whose value is {?} in this case. The corresponding p-value is small (large), so we do not reject (reject) the hypothesis of association at any reasonable significance level.

Reason| 是否能用边缘分布来推断 (t-test): [1] Case I: design matrix **X** orthogonal (or 其他方式得到 parameter independent), it's possible to use the marginal distribution of $\underline{\widehat{\beta}}$ to make inference. [2] Case II: **X** not orthogonal (or 其他方式得到 parameter dependent), it's not possible to use the marginal distribution of $\underline{\widehat{\beta}}$ to make inference.

Method| 是否能用边缘分布来推断 (t-test): 如果不能使用: Using 4.5 后半段内容.

***Not Exam|centered simple linear regression**: 将 x 变为 $x' = x - \overline{x}$, ⇒ Design matrix will be orthogonal.